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(54) **SYSTEM AND METHOD FOR ENHANCING PROPELLER IMAGE QUALITY BY UTILIZING MULTI-LEVEL DENOISING**

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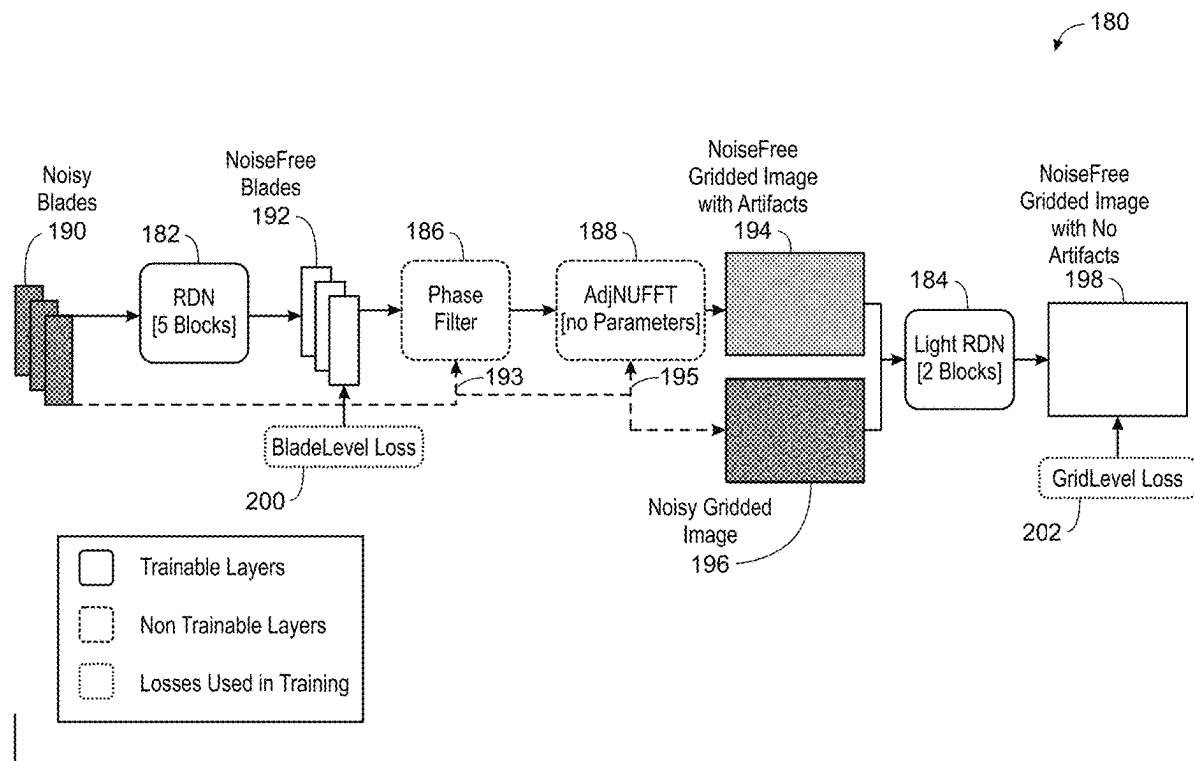
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(57)

ABSTRACT

A system and method for improving image quality of periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) imaging include acquiring a plurality of blades of k-space data of a region of interest in a rotational manner around a center of k-space via a magnetic resonance imaging (MRI) scanner from a coil during a PROPELLER sequence, wherein each blade of the plurality of blades of k-space data includes a plurality of parallel phase encoding lines sampled in a phase encoding order. The system and method also include utilizing a deep learning-based multi-level denoising network to denoise each blade of the plurality of blades in an image domain to generate a plurality of denoised blades, to utilize a PROPELLER reconstruction algorithm to generate a denoised-gridded image from the plurality of denoised blades, and to remove individual-based denoising-induced artifacts from the denoised-gridded image to generate a denoised, artifact-free gridded image.



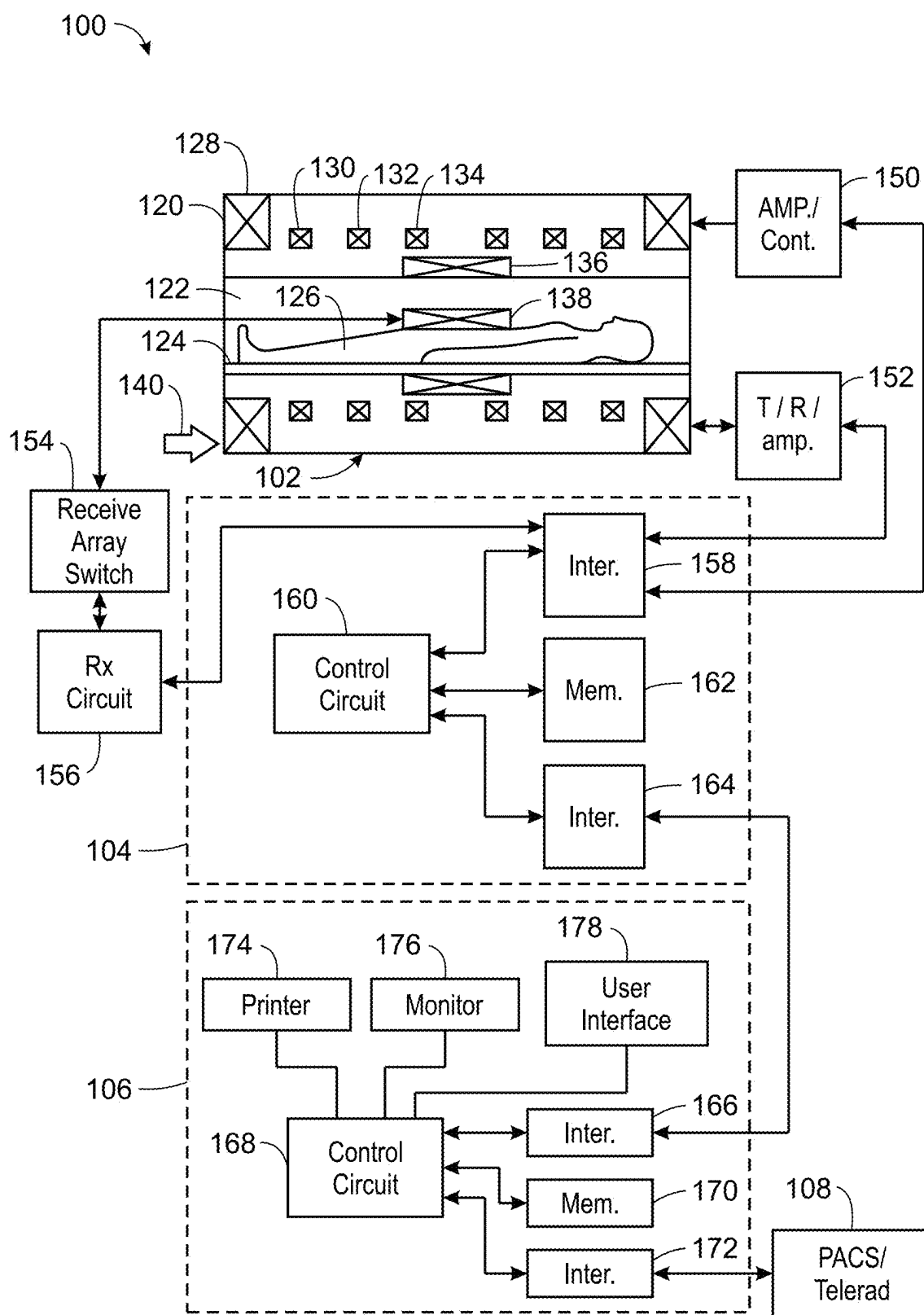


FIG. 1

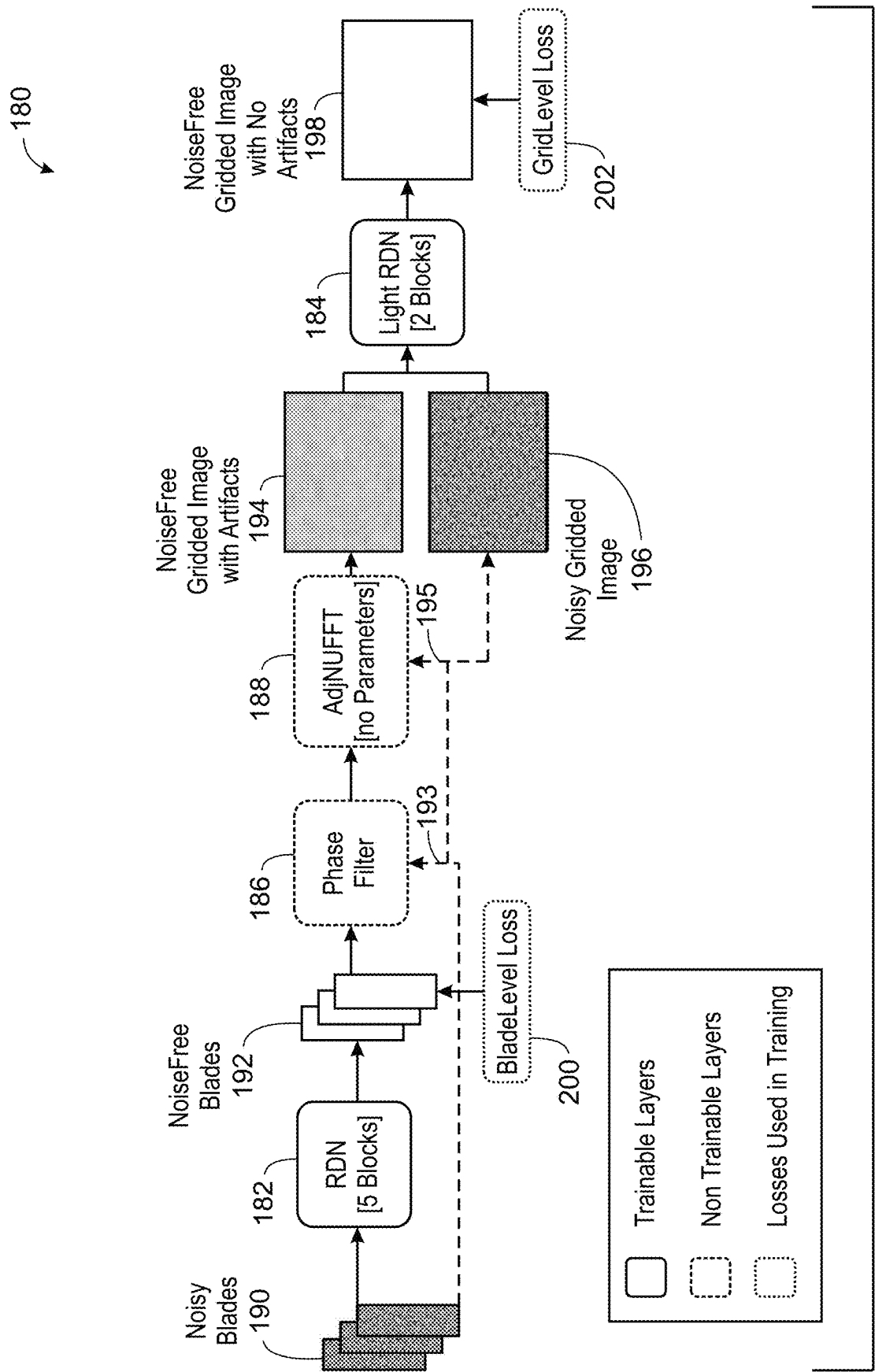


FIG. 2

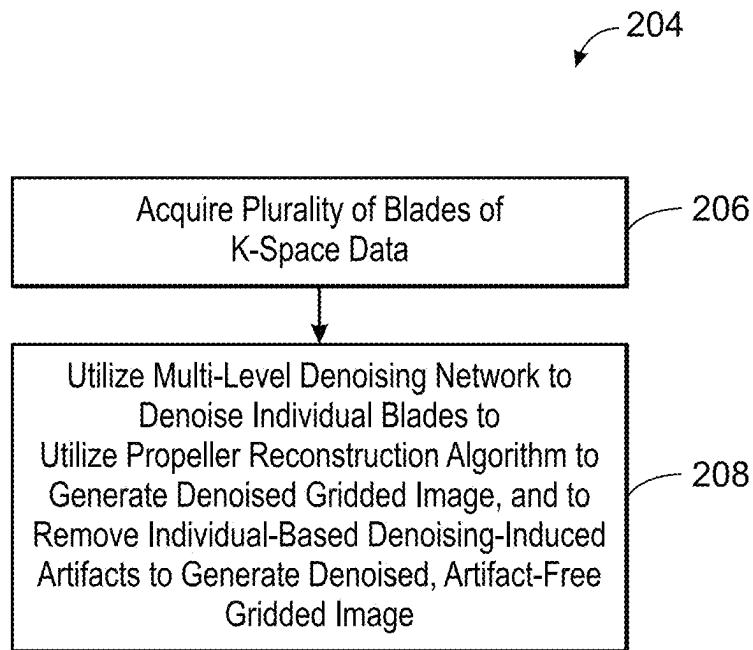


FIG. 3

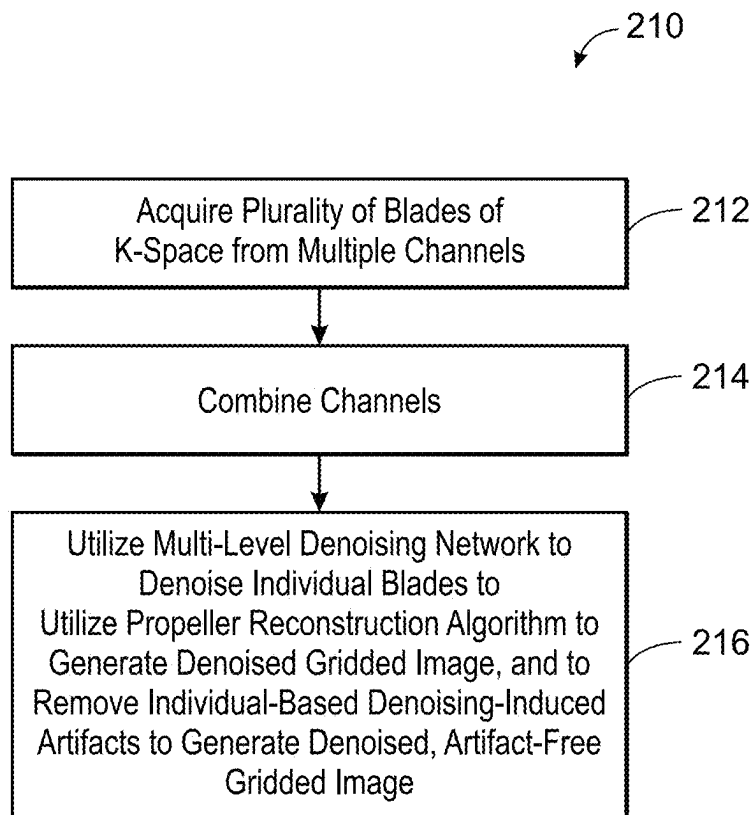


FIG. 4

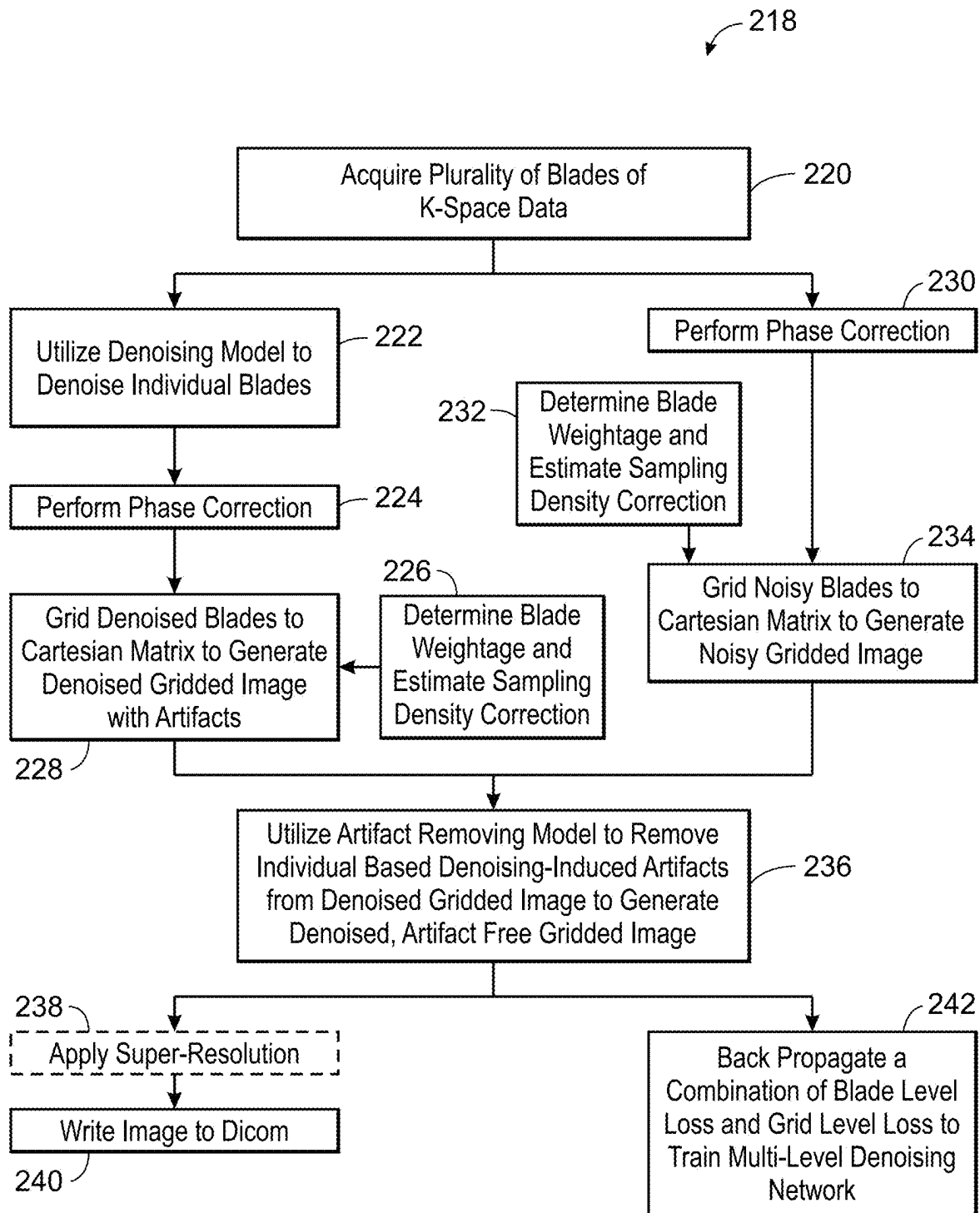


FIG. 5

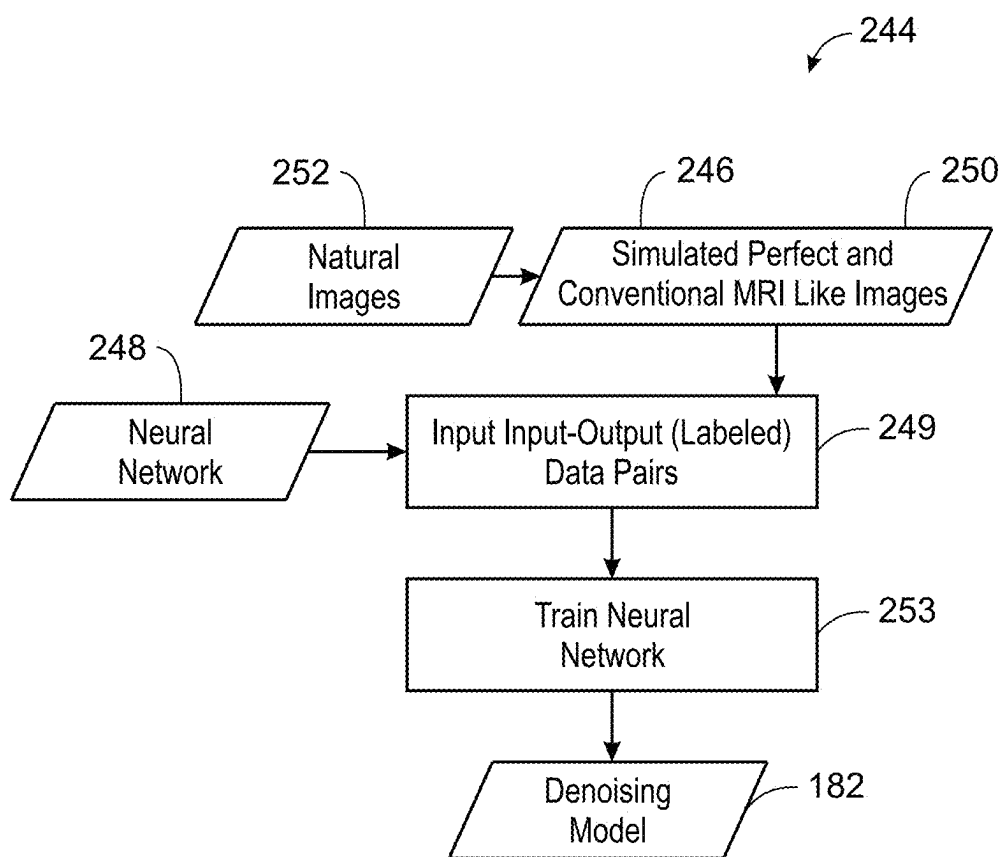


FIG. 6

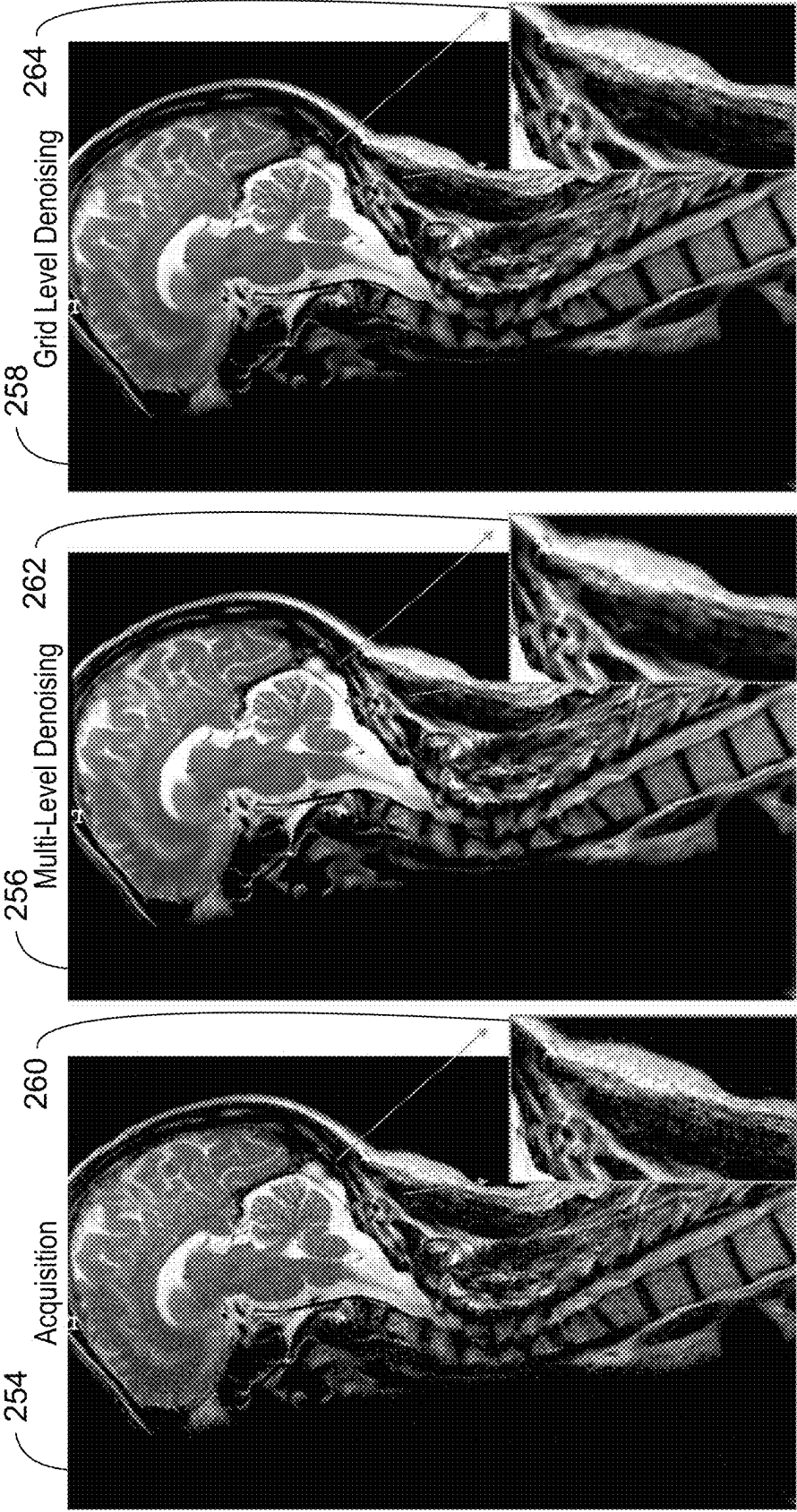


FIG. 7

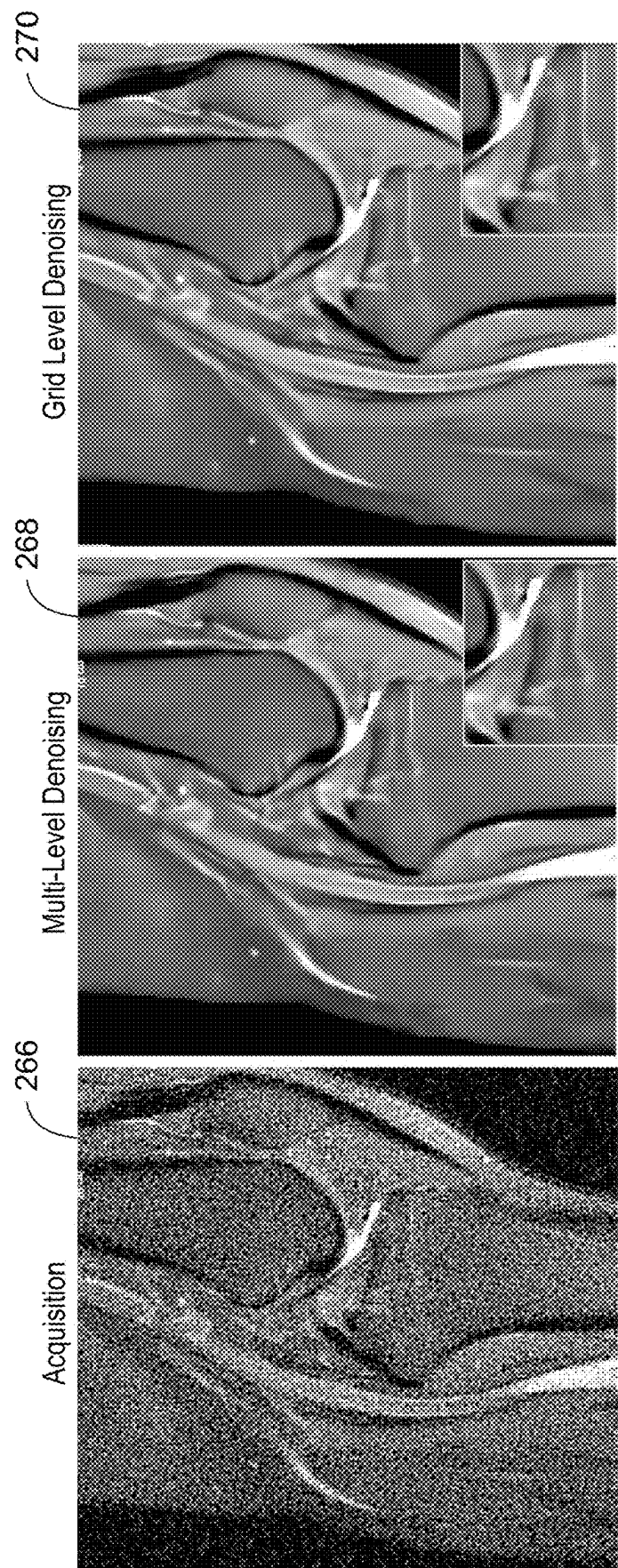


FIG. 8

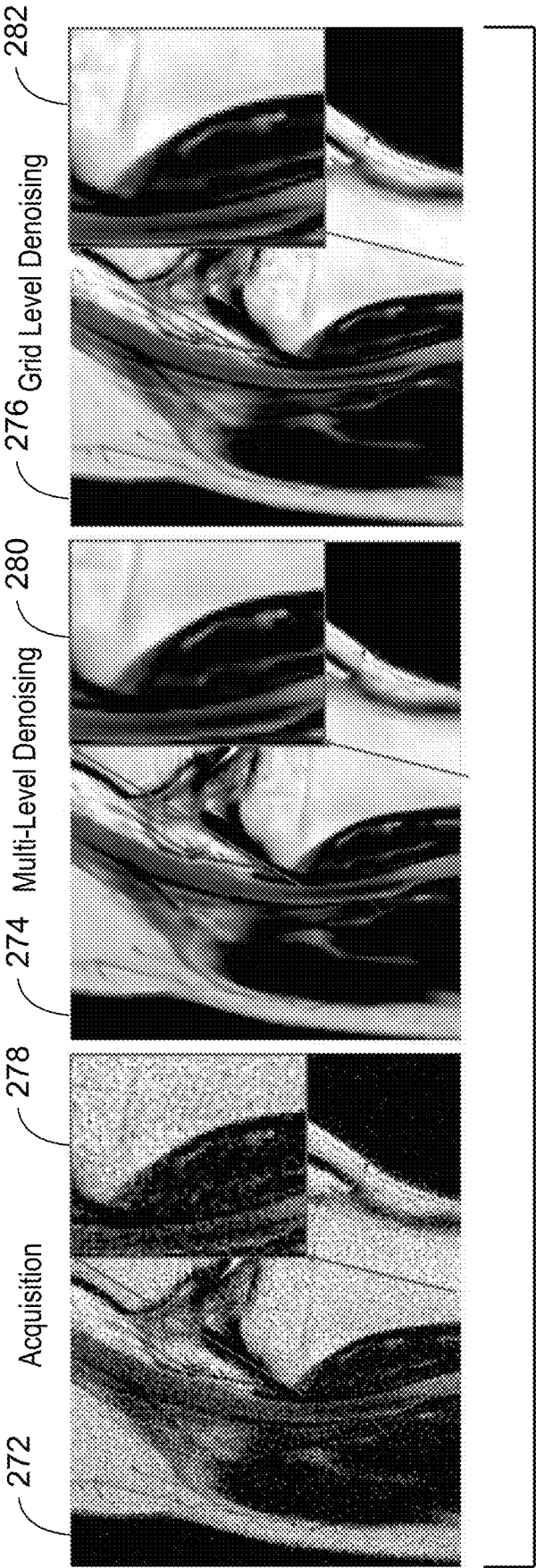


FIG. 9

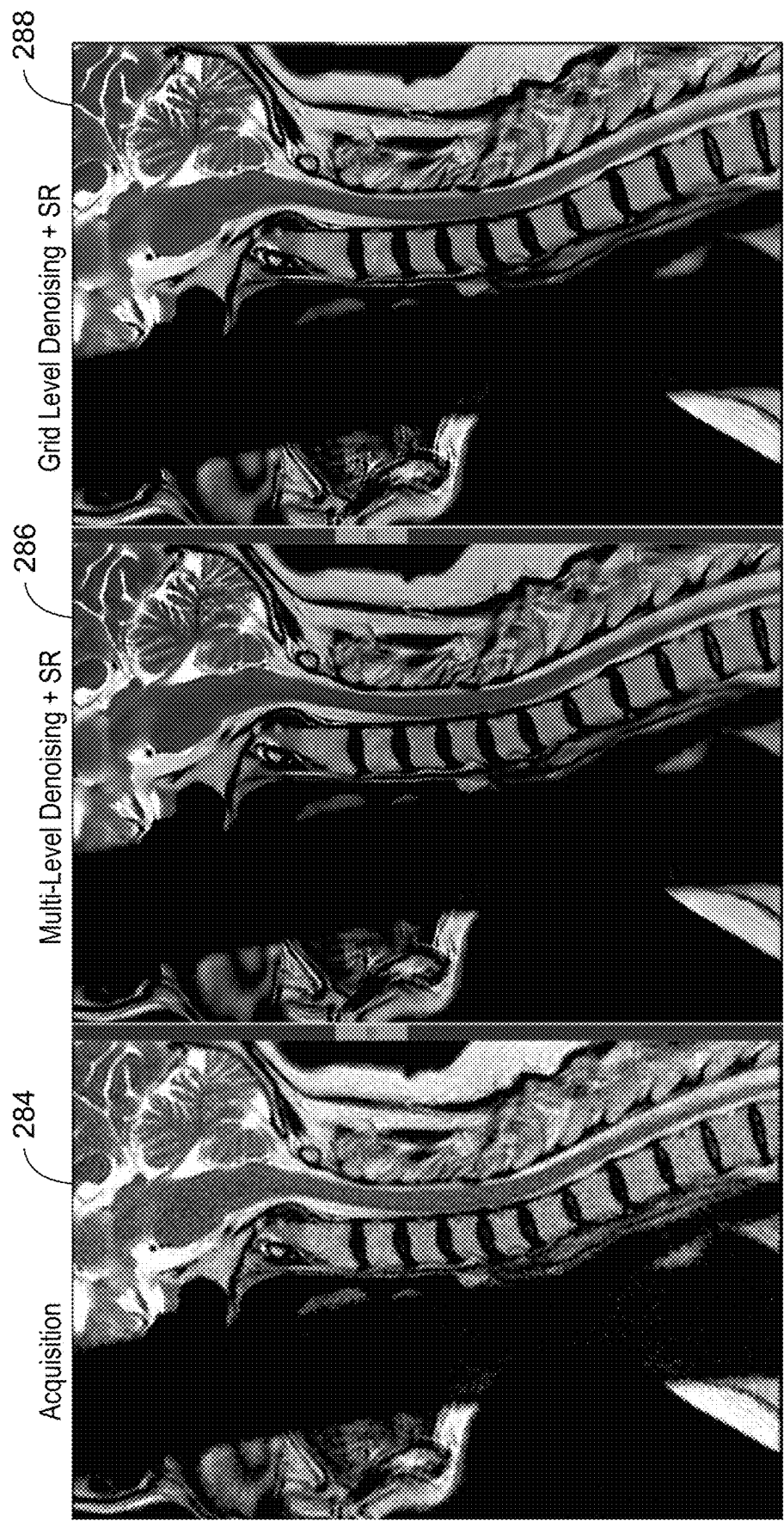


FIG. 10

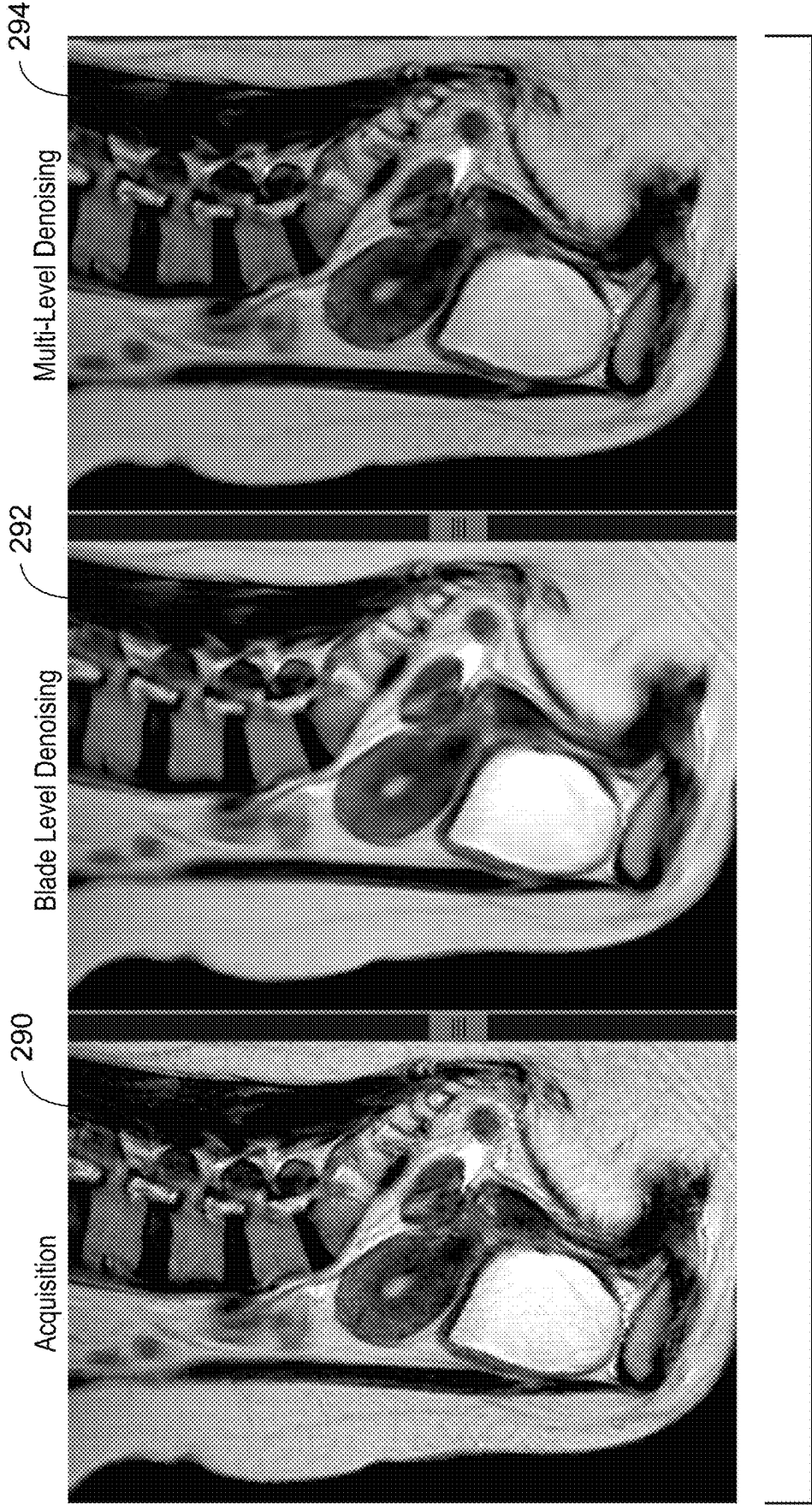


FIG. 11

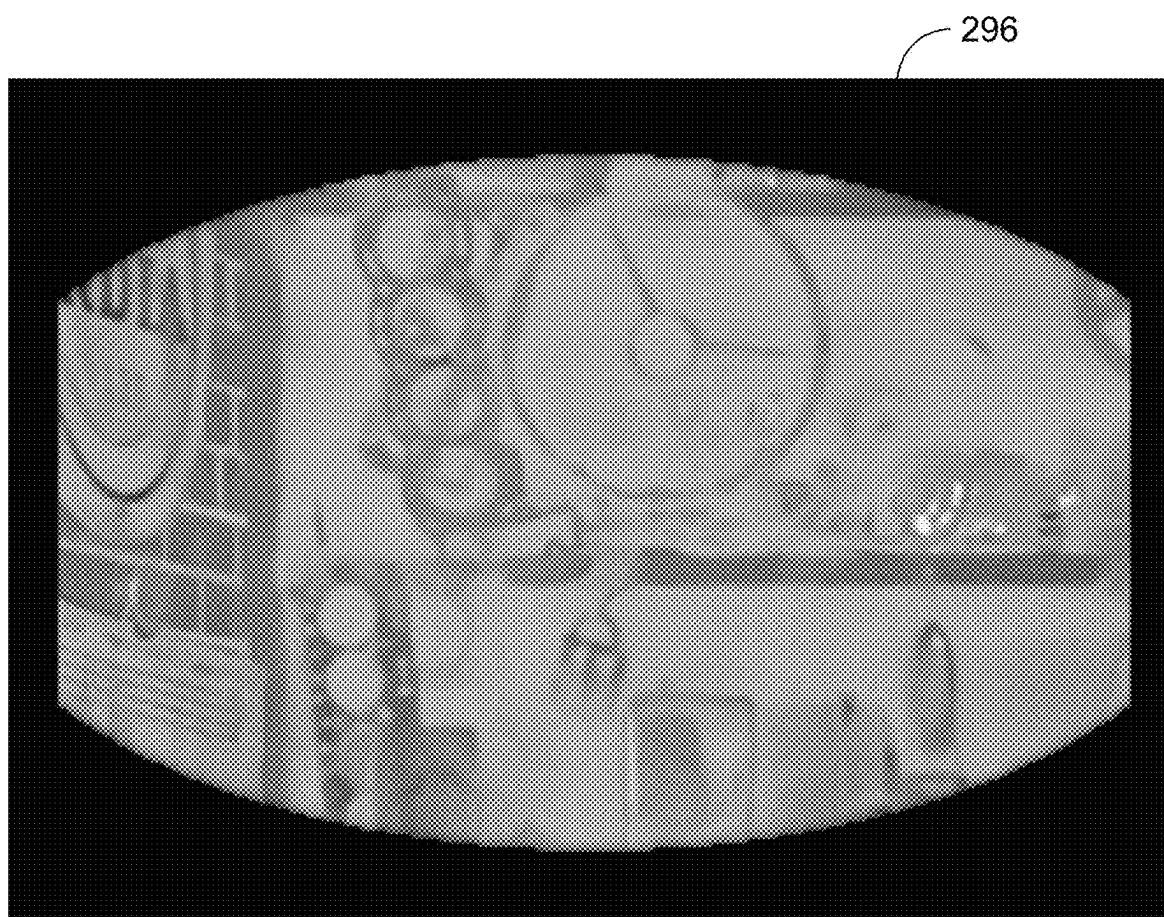


FIG. 12

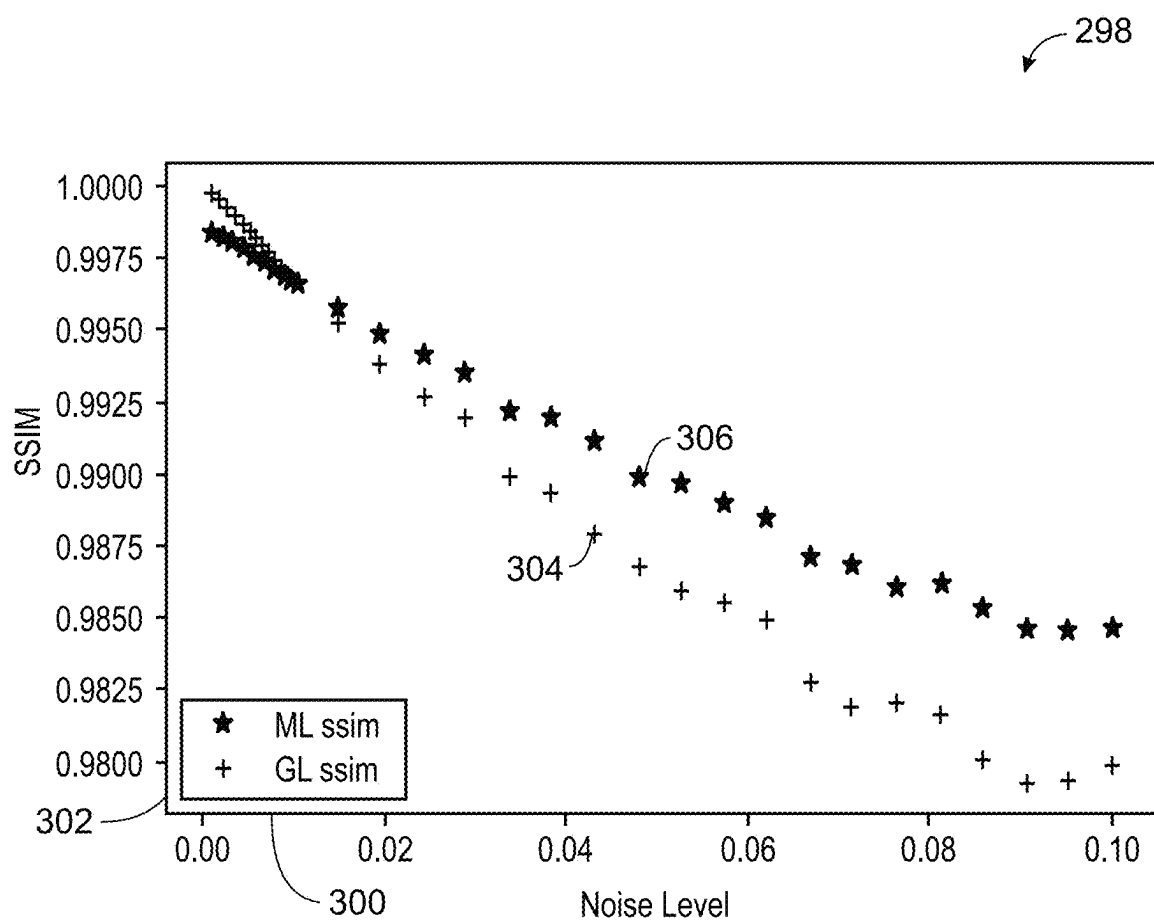


FIG. 13

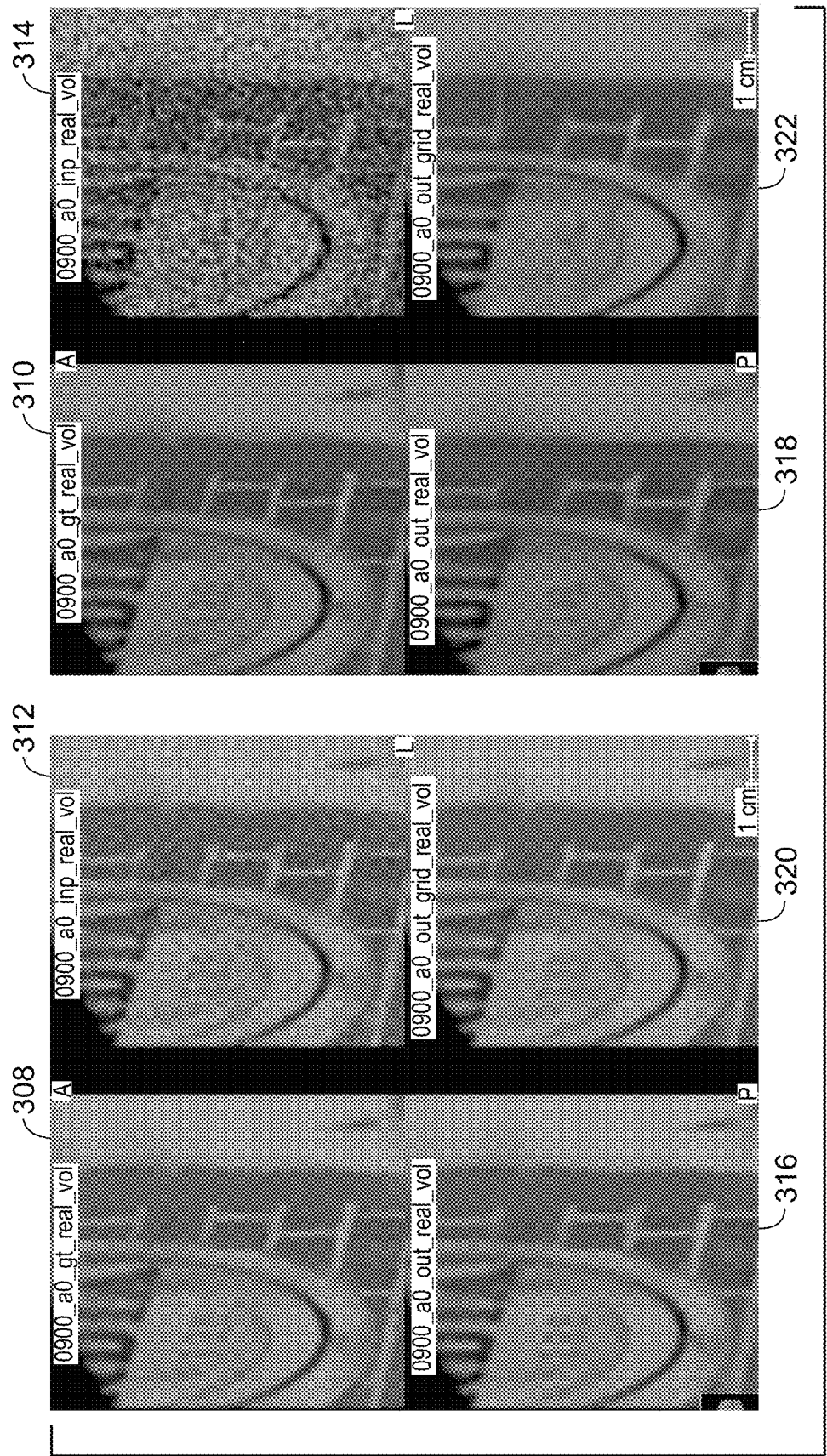


FIG. 14

SYSTEM AND METHOD FOR ENHANCING PROPELLER IMAGE QUALITY BY UTILIZING MULTI-LEVEL DENOISING

BACKGROUND

[0001] The subject matter disclosed herein relates to medical imaging and, more particularly, to a system and a method for enhancing propeller image quality by utilizing multi-level denoising.

[0002] Non-invasive imaging technologies allow images of the internal structures or features of a patient/object to be obtained without performing an invasive procedure on the patient/object. In particular, such non-invasive imaging technologies rely on various physical principles (such as the differential transmission of X-rays through a target volume, the reflection of acoustic waves within the volume, the paramagnetic properties of different tissues and materials within the volume, the breakdown of targeted radionuclides within the body, and so forth) to acquire data and to construct images or otherwise represent the observed internal features of the patient/object.

[0003] During magnetic resonance imaging (MRI), when a substance such as human tissue is subjected to a uniform magnetic field (polarizing field B_0), the individual magnetic moments of the spins in the tissue attempt to align with this polarizing field, but precess about it in random order at their characteristic Larmor frequency. If the substance, or tissue, is subjected to a magnetic field (excitation field B_1) which is in the x-y plane and which is near the Larmor frequency, the net aligned moment, or “longitudinal magnetization”, M_z , may be rotated, or “tipped”, into the x-y plane to produce a net transverse magnetic moment, M_r . A signal is emitted by the excited spins after the excitation signal B_1 is terminated and this signal may be received and processed to form an image.

[0004] When utilizing these signals to produce images, magnetic field gradients (G_x , G_y , and G_z) are employed. Typically, the region to be imaged is scanned by a sequence of measurement cycles in which these gradient fields vary according to the particular localization method being used. The resulting set of received nuclear magnetic resonance (NMR) signals are digitized and processed to reconstruct the image using one of many well-known reconstruction techniques.

[0005] Periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) is an MR technique which provides high resolution magnetic resonance imaging with reduced motion artifacts by providing the capability to remove motion affected blades and by over-sampling the low spatial frequencies. While the noise is Gaussian in each blade, after gridding the blades into a Cartesian matrix, the noise becomes colored. Attempts at denoising colored noise are reported to produce blurriness and degraded image quality. The denoising performance becomes drastically worse in the case of low signal-to-noise ratio data.

BRIEF DESCRIPTION

[0006] A summary of certain embodiments disclosed herein is set forth below. It should be understood that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure.

Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

[0007] In one embodiment, computer-implemented method for improving image quality of periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) imaging is provided. The computer-implemented method includes acquiring, via a processor, a plurality of blades of k-space data of a region of interest in a rotational manner around a center of k-space via a magnetic resonance imaging (MRI) scanner from a coil during a PROPELLER sequence, wherein each blade of the plurality of blades of k-space data includes a plurality of parallel phase encoding lines sampled in a phase encoding order. The computer-implemented method also includes utilizing, via the processor, a deep learning-based multi-level denoising network to denoise each blade of the plurality of blades in an image domain to generate a plurality of denoised blades, to utilize a PROPELLER reconstruction algorithm to generate a denoised-gridded image from the plurality of denoised blades, and to remove individual-based denoising-induced artifacts from the denoised-gridded image to generate a denoised, artifact-free gridded image.

[0008] In another embodiment, a system for improving image quality of periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) imaging is provided. The system includes a memory encoding processor-executable routines. The system also includes a processor configured to access the memory and to execute the processor-executable routines, wherein the processor-executable routines, when executed by the processor, cause the processor to perform actions. The actions include acquiring a plurality of blades of k-space data of a region of interest in a rotational manner around a center of k-space via a magnetic resonance imaging (MRI) scanner from a coil during a PROPELLER sequence, wherein each blade of the plurality of blades of k-space data includes a plurality of parallel phase encoding lines sampled in a phase encoding order. The actions also include utilizing a deep learning-based multi-level denoising network to denoise each blade of the plurality of blades in an image domain to generate a plurality of denoised blades, to utilize a PROPELLER reconstruction algorithm to generate a denoised-gridded image from the plurality of denoised blades, and to remove individual-based denoising-induced artifacts from the denoised-gridded image to generate a denoised, artifact-free gridded image.

[0009] In a further embodiment, a non-transitory computer-readable medium, the computer-readable medium including processor-executable code that when executed by a processor, causes the processor to perform actions. The actions include acquiring a plurality of blades of k-space data of a region of interest in a rotational manner around a center of k-space via a magnetic resonance imaging (MRI) scanner from a coil during a PROPELLER sequence, wherein each blade of the plurality of blades of k-space data includes a plurality of parallel phase encoding lines sampled in a phase encoding order. The actions also include utilizing a deep learning-based multi-level denoising network to denoise each blade of the plurality of blades in an image domain to generate a plurality of denoised blades, to utilize a PROPELLER reconstruction algorithm to generate a denoised-gridded image from the plurality of denoised blades, and to remove individual-based denoising-induced

artifacts from the denoised-gridded image to generate a denoised, artifact-free gridded image.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] These and other features, aspects, and advantages of the present subject matter will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

[0011] FIG. 1 illustrates an embodiment of a magnetic resonance imaging (MRI) system suitable for use with the disclosed technique;

[0012] FIG. 2 illustrates a schematic diagram illustrating a deep learning-based multi-level denoising network configured to improve image quality of PROPELLER imaging, in accordance with aspects of the present disclosure;

[0013] FIG. 3 illustrates a flow chart of a method for improving image quality of periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) imaging (e.g., utilizing data from a single channel), in accordance with aspects of the present disclosure;

[0014] FIG. 4 illustrates a flow chart of a method for improving image quality of periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) imaging (e.g., utilizing data from multiple channels), in accordance with aspects of the present disclosure;

[0015] FIG. 5 illustrates a flow chart of a method for improving image quality of periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) imaging (e.g., with details of PROPELLER reconstruction), in accordance with aspects of the present disclosure;

[0016] FIG. 6 illustrates a flow chart of a method for training a deep learning-based denoising network, in accordance with aspects of the present disclosure;

[0017] FIG. 7 depicts periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) images of a spine comparing multi-level denoising and grid level denoising, in accordance with aspects of the present disclosure;

[0018] FIG. 8 depicts PROPELLER images of a knee comparing multi-level denoising and grid level denoising, in accordance with aspects of the present disclosure;

[0019] FIG. 9 depicts additional PROPELLER images of a knee comparing multi-level denoising and grid level denoising, in accordance with aspects of the present disclosure;

[0020] FIG. 10 depicts PROPELLER images of a spine comparing multi-level denoising and grid level denoising with super-resolution applied, in accordance with aspects of the present disclosure;

[0021] FIG. 11 depicts PROPELLER images of a pelvis comparing blade level and multi-level denoising, in accordance with aspects of the present disclosure;

[0022] FIG. 12 depicts an example of a PROPELLER ground truth image of a natural image, in accordance with aspects of the present disclosure;

[0023] FIG. 13 depicts a graph analyzing performance of grid level denoising relative to multi-level denoising on a natural image, in accordance with aspects of the present disclosure; and

[0024] FIG. 14 depicts examples of images from the analysis in FIG. 13, in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

[0025] One or more specific embodiments will be described below. In an effort to provide a concise description of these embodiments, not all features of an actual implementation are described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

[0026] When introducing elements of various embodiments of the present subject matter, the articles "a," "an," "the," and "said" are intended to mean that there are one or more of the elements. The terms "comprising," "including," and "having" are intended to be inclusive and mean that there may be additional elements other than the listed elements. Furthermore, any numerical examples in the following discussion are intended to be non-limiting, and thus additional numerical values, ranges, and percentages are within the scope of the disclosed embodiments.

[0027] While aspects of the following discussion are provided in the context of medical imaging, it should be appreciated that the disclosed techniques are not limited to such medical contexts. Indeed, the provision of examples and explanations in such a medical context is only to facilitate explanation by providing instances of real-world implementations and applications. However, the disclosed techniques may also be utilized in other contexts, such as image reconstruction for non-destructive inspection of manufactured parts or goods (i.e., quality control or quality review applications), and/or the non-invasive inspection of packages, boxes, luggage, and so forth (i.e., security or screening applications). In general, the disclosed techniques may be useful in any imaging or screening context or image processing or photography field where a set or type of acquired data undergoes a reconstruction process to generate an image or volume.

[0028] Deep learning (DL) approaches discussed herein may be based on artificial neural networks, and may therefore encompass one or more of deep neural networks, fully connected networks, convolutional neural networks (CNNs), unrolled neural networks, perceptrons, encoders-decoders, recurrent networks, transformer networks, wavelet filter banks, u-nets, general adversarial networks (GANs), dense neural networks (e.g., residual dense networks (RDNs), or other neural network architectures. The neural networks may include shortcuts, activations, batch-normalization layers, and/or other features. These techniques are referred to herein as DL techniques, though this terminology may also be used specifically in reference to the use of deep neural networks, which is a neural network having a plurality of layers.

[0029] As discussed herein, DL techniques (which may also be known as deep machine learning, hierarchical learning, or deep structured learning) are a branch of machine learning techniques that employ mathematical representations of data and artificial neural networks for learning and processing such representations. By way of example, DL

approaches may be characterized by their use of one or more algorithms to extract or model high level abstractions of a type of data-of-interest. This may be accomplished using one or more processing layers, with each layer typically corresponding to a different level of abstraction and, therefore potentially employing or utilizing different aspects of the initial data or outputs of a preceding layer (i.e., a hierarchy or cascade of layers) as the target of the processes or algorithms of a given layer. In an image processing or reconstruction context, this may be characterized as different layers corresponding to the different feature levels or resolution in the data. In general, the processing from one representation space to the next-level representation space can be considered as one 'stage' of the process. Each stage of the process can be performed by separate neural networks or by different parts of one larger neural network.

[0030] In magnetic resonance imaging, noise is Gaussian in the acquired complex k-space as well as in the complex image. Periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) is an MR technique which provides high resolution magnetic resonance imaging with reduced motion artifacts by providing the capability to remove motion affected blades and by over-sampling the low spatial frequencies. In PROPELLER, overlapping blades are acquired rotated around the k-space. Thus, the noise remains Gaussian in the blades. However, to obtain the end image, the blades need to be phase corrected and then gridded into uniform k-space coordinates (i.e., a Cartesian matrix), which causes the noise to become colored. Typically, denoising is performed after generating the complex image (when the noise is colored). But denoising models have been observed to perform better with Gaussian noise than colored noise. Attempts at denoising colored noise are reported to produce blurriness and degraded image quality. The denoising performance becomes drastically worse in the case of low signal-to-noise ratio data.

[0031] The present disclosure provides systems and methods for improving image quality of periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) imaging. In particular, the present disclosure provides for denoising individual blades in an image domain prior to phase correcting and gridding the blades (where the noise is still Gaussian). In very low signal-to-noise ratio cases, a deep learning-based model utilized to individually denoise the individual blades tends to act aggressively to remove noise, thereby, introducing blur. But, in each blade, the image is rotated differently and the deep learning-based model filters act along rows and columns only. This implies that the blur introduced by the deep learning-based model on each blade is along a relatively different angle. This blur when gridded manifests as streaks and small structures are prone to completely dissipate. This effect becomes more pronounced as the blade width becomes narrower. Thus, denoising individual blades may introduce artifacts that may arise after gridding together of independent deep learning artifacts such as blurring and half pixel shifts introduced in each blade. These artifacts may end up in structures streaking over after gridding.

[0032] In order to take advantage of the benefits of blade level denoising while also avoiding the downside, the disclosed embodiments provide systems and methods for multi-level denoising. In multi-level denoising, a deep learning-based denoising model (e.g., residual dense network model of 5 blocks) pretrained to remove Gaussian noise is first

utilized. This is followed with parameter-less phase correction and adjoint non-uniform fast Fourier transform blocks to correct for different phase in each of the blades and to grid the blades into a Cartesian grid, respectively. A deep learning-based artifact removing model (e.g., light or lightweight residual dense network model having only 2 blocks) is then utilized. The entire set-up (i.e., deep learning-based multi-level denoising network) is then trained end to end to perform better denoising with less artifacts. In particular, both the deep learning-based denoising model and the deep learning-based artifact removing model are trained with backpropagated blade level loss and loss obtained after gridding of blades (i.e., grid level loss). The deep learning-based artifact removing model handles the blade level denoising artifacts that remain and smaller structures that are dissipated. A gridded image (obtained by gridding denoised blades) and a noisy gridded image (obtained after gridding the acquired noisy blades without denoising) are inputted into the deep learning-based artifact removing model to generate a denoised gridded image that is free of the blade level denoising model introduced artifacts. The deep learning-based artifact removing model is trained against a ground truth noise-free gridded image. The deep learning-based multi-level denoising network is configured to perform very clean denoising with high fidelity. By virtue of the training process, streaks are also removed. The training simulation can be extended to include motion robust gridding, where different blades are weighed differently before gridding and some blades may even be dropped. This causes non-uniform k-space which gives rise to streaks and the deep learning-based artifact removing model would become equipped to handles these streaks as well. The deep learning-based multi-level denoising network provides an end to end trainable deep learning framework completely immersive into PROPELLER reconstruction that is configured to achieve denoising with high fidelity and minimal artifacts.

[0033] The disclosed systems and methods include acquiring a plurality of blades of k-space data of a region of interest in a rotational manner around a center of k-space via a magnetic resonance imaging (MRI) scanner from a coil during a PROPELLER sequence, wherein each blade of the plurality of blades of k-space data includes a plurality of parallel phase encoding lines sampled in a phase encoding order. The disclosed systems and methods also include utilizing a deep learning-based multi-level denoising network to denoise each blade of the plurality of blades in an image domain to generate a plurality of denoised blades, to utilize a PROPELLER reconstruction algorithm to generate a denoised-gridded image from the plurality of denoised blades, and to remove individual-based denoising-induced artifacts from the denoised-gridded image to generate a denoised, artifact-free gridded image.

[0034] In certain embodiments, the deep learning-based multi-level denoising network includes a deep learning-based denoising model to denoise each blade of the plurality of blades in the image domain to generate the plurality of denoised blades and a deep learning-based artifact removing model to remove the individual-based denoising-induced artifacts from the denoised-gridded image to generate the denoised, artifact-free gridded image, and wherein the PROPELLER reconstruction algorithm includes adjoint non-uniform fast Fourier transform blocks configured to grid the plurality of denoised blades into a Cartesian grid. In certain embodiments, the deep learning-based multi-level denoising

network includes a phase filter disposed between the deep learning-based denoising model and the adjoint non-uniform fast Fourier transform blocks that is configured to perform parameter-less phase correction to correct for different phase in each of the blades. In certain embodiments, the deep learning-based artifact removing model has fewer parameters than the deep learning-based denoising model.

[0035] In certain embodiments, the disclosed systems and methods include backpropagating a combination of both blade level loss and grid level loss to train the deep learning-based multi-level denoising network from end to end. In certain embodiments, the combination of both the blade level loss and the grid level loss are backpropagated to train both the deep learning-based denoising model and the deep learning-based artifact removing model.

[0036] In certain embodiments, the disclosed systems and methods include utilizing the PROPELLER reconstruction algorithm to generate a noisy gridded image from the plurality of blades that have not been denoised. In certain embodiments, the disclosed systems and methods include inputting both the denoised-gridded image and the noisy gridded image into the deep learning-based artifact removing model, wherein the deep learning-based artifact removing model utilizes both the denoised-gridded image and the noisy gridded image to generate the denoised, artifact-free gridded image. In certain embodiments, the disclosed systems and methods include applying super-resolution to the denoised, artifact-free gridded image to generate a higher resolution denoised, artifact-free gridded image that is further denoised. In certain embodiments, the denoised, artifact-free gridded image may be subject to other post-processing. In certain embodiments, the disclosed systems and methods further include writing the denoised, artifact-free gridded image to a DICOM file (e.g., for viewing or storage).

[0037] In certain embodiments, the plurality of blades of k-space data is acquired from a single channel of the coil. In certain embodiments, the plurality of blades of k-space data is acquired from a plurality of channels of the coil. In these embodiments, the disclosed systems and methods include combining the plurality of blades of k-space data acquired from the plurality of channels prior to utilizing the deep learning-based denoising network to denoise each blade of the plurality of blades in the image domain.

[0038] PROPELLER is generally used to suppress motion. Also, to suppress motion PROPELLER might be accelerated but this decreases the signal-to-noise ratio. Further, if the imaging system is further de-rated (with regard to number of coils and field-strength utilized), the signal-to-noise ratio falls below diagnosable quality. The disclosed systems and methods improve image quality in with regard to both signal-to-noise ratio and sharpness.

[0039] With the preceding in mind, FIG. 1 a magnetic resonance imaging (MRI) system **100** is illustrated schematically as including a scanner **102**, scanner control circuitry **104**, and system control circuitry **106**. According to the embodiments described herein, the magnetic resonance imaging system **100** is generally configured to perform magnetic resonance imaging.

[0040] System **100** additionally includes remote access and storage systems or devices such as picture archiving and communication systems (PACS) **108**, or other devices such as teleradiology equipment so that data acquired by the system **100** may be accessed on- or off-site. In this way, MR

data may be acquired, followed by on- or off-site processing and evaluation. While the magnetic resonance imaging system **100** may include any suitable scanner or detector, in the illustrated embodiment, the system **100** includes a full body scanner **102** having a housing **120** through which a bore **122** is formed. A table **124** is moveable into the bore **122** to permit a patient **126** (e.g., subject) to be positioned therein for imaging selected anatomy within the patient.

[0041] Scanner **102** includes a series of associated coils for producing controlled magnetic fields for exciting the gyromagnetic material within the anatomy of the patient being imaged. Specifically, a primary magnet coil **128** is provided for generating a primary magnetic field, B_0 , which is generally aligned with the bore **122**. A series of gradient coils **130**, **132**, and **134** permit controlled magnetic gradient fields to be generated for positional encoding of certain gyromagnetic nuclei within the patient **126** during examination sequences. A radio frequency (RF) coil **136** (e.g., RF transmit coil) is configured to generate radio frequency pulses for exciting the certain gyromagnetic nuclei within the patient. In addition to the coils that may be local to the scanner **102**, the system **100** also includes a set of receiving coils or RF receiving coils **138** (e.g., an array of coils) configured for placement proximal (e.g., against) to the patient **126**. As an example, the receiving coils **138** can include cervical/thoracic/lumbar (CTL) coils, head coils, single-sided spine coils, and so forth. Generally, the receiving coils **138** are placed close to or on top of the patient **126** so as to receive the weak RF signals (weak relative to the transmitted pulses generated by the scanner coils) that are generated by certain gyromagnetic nuclei within the patient **126** as they return to their relaxed state.

[0042] The various coils of system **100** are controlled by external circuitry to generate the desired field and pulses, and to read emissions from the gyromagnetic material in a controlled manner. In the illustrated embodiment, a main power supply **140** provides power to the primary field coil **128** to generate the primary magnetic field, B_0 . A power input (e.g., power from a utility or grid), a power distribution unit (PDU), a power supply (PS), and a driver circuit **150** may together provide power to pulse the gradient field coils **130**, **132**, and **134**. The driver circuit **150** may include amplification and control circuitry for supplying current to the coils as defined by digitized pulse sequences output by the scanner control circuitry **104**.

[0043] Another control circuit **152** is provided for regulating operation of the RF coil **136**. Circuit **152** includes a switching device for alternating between the active and inactive modes of operation, wherein the RF coil **136** transmits and does not transmit signals, respectively. Circuit **152** also includes amplification circuitry configured to generate the RF pulses. Similarly, the receiving coils **138** are connected to switch **154**, which is capable of switching the receiving coils **138** between receiving and non-receiving modes. Thus, the receiving coils **138** resonate with the RF signals produced by relaxing gyromagnetic nuclei from within the patient **126** while in the receiving mode, and they do not resonate with RF energy from the transmitting coils (i.e., coil **136**) so as to prevent undesirable operation while in the non-receiving mode. Additionally, a receiving circuit **156** is configured to receive the data detected by the receiving coils **138** and may include one or more multiplexing and/or amplification circuits.

[0044] It should be noted that while the scanner **102** and the control/amplification circuitry described above are illustrated as being coupled by a single line, many such lines may be present in an actual instantiation. For example, separate lines may be used for control, data communication, power transmission, and so on. Further, suitable hardware may be disposed along each type of line for the proper handling of the data and current/voltage. Indeed, various filters, digitizers, and processors may be disposed between the scanner and either or both of the scanner and system control circuitry **104**, **106**.

[0045] As illustrated, scanner control circuitry **104** includes an interface circuit **158**, which outputs signals for driving the gradient field coils and the RF coil and for receiving the data representative of the magnetic resonance signals produced in examination sequences. The interface circuit **158** is coupled to a control and analysis circuit **160**. The control and analysis circuit **160** executes the commands for driving the circuit **150** and circuit **152** based on defined protocols selected via system control circuit **106**.

[0046] Control and analysis circuit **160** also serves to receive the magnetic resonance signals and performs subsequent processing before transmitting the data to system control circuit **106**. Scanner control circuit **104** also includes one or more memory circuits **162**, which store configuration parameters, pulse sequence descriptions, examination results, and so forth, during operation.

[0047] Interface circuit **164** is coupled to the control and analysis circuit **160** for exchanging data between scanner control circuitry **104** and system control circuitry **106**. In certain embodiments, the control and analysis circuit **160**, while illustrated as a single unit, may include one or more hardware devices. The system control circuit **106** includes an interface circuit **166**, which receives data from the scanner control circuitry **104** and transmits data and commands back to the scanner control circuitry **104**. The control and analysis circuit **168** may include a CPU in a multi-purpose or application specific computer or workstation. Control and analysis circuit **168** is coupled to a memory circuit **170** to store programming code for operation of the magnetic resonance imaging system **100** and to store the processed image data for later reconstruction, display and transmission. The programming code may execute one or more algorithms that, when executed by a processor, are configured to perform reconstruction of acquired data as described below. In certain embodiments, the memory circuit **170** may store one or more neural networks for processing and/or reconstruction of acquired data (e.g., deep learning-based multi-level denoising network) as described below. In certain embodiments, image reconstruction may occur on a separate computing device having processing circuitry and memory circuitry.

[0048] A processing component (e.g., a microprocessor or processing circuitry) and a memory of the magnetic resonance imaging system **100**, such as may be present in scanner control circuitry **104** and/or system control circuitry **106**, may be used to execute stored software code, instructions, or routines for acquiring and processing the MR data. The term “code” or “software code” used herein refers to any instructions or set of instructions that control the magnetic resonance imaging system **100**. The code or software code may exist in a computer-executable form, such as machine code, which is the set of instructions and data directly executed by the processing component of the scan-

ner control circuitry **104** and/or system control circuitry **106**, human-understandable form, such as source code, which may be compiled in order to be executed by the processing component of the scanner control circuitry **104** and/or system control circuitry **106**, or an intermediate form, such as object code, which is produced by a compiler. In some embodiments, the magnetic resonance imaging system **100** may include a plurality of controllers.

[0049] As an example, the memory may store processor-executable software code or instructions (e.g., firmware or software), which are tangibly stored on a non-transitory computer readable medium. Additionally or alternatively, the memory **46** may store data. As an example, the memory may include a volatile memory, such as random access memory (RAM), and/or a nonvolatile memory, such as read-only memory (ROM), flash memory, a hard drive, or any other suitable optical, magnetic, or solid-state storage medium, or a combination thereof. Furthermore, processing component may include multiple microprocessors, one or more “general-purpose” microprocessors, one or more special-purpose microprocessors, and/or one or more application specific integrated circuits (ASICs), or some combination thereof. For example, the processing component may include one or more reduced instruction set (RISC) or complex instruction set (CISC) processors. The processing component may include multiple processors, and/or the memory may include multiple memory devices.

[0050] In certain embodiments, the processing component may be configured to acquire a plurality of blades of k-space data of a region of interest in a rotational manner around a center of k-space via a magnetic resonance imaging (MRI) scanner from a coil during a PROPELLER sequence, wherein each blade of the plurality of blades of k-space data includes a plurality of parallel phase encoding lines sampled in a phase encoding order. The processing component may also be configured to utilizing a deep learning-based multi-level denoising network to denoise each blade of the plurality of blades in an image domain to generate a plurality of denoised blades, to utilize a PROPELLER reconstruction algorithm to generate a denoised-gridded image from the plurality of denoised blades, and to remove individual-based denoising-induced artifacts from the denoised-gridded image to generate a denoised, artifact-free gridded image.

[0051] The deep learning-based multi-level denoising network includes a deep learning-based denoising model to denoise each blade of the plurality of blades in an image domain to generate the plurality of denoised blades and a deep learning-based artifact removing model to remove the individual-based denoising-induced artifacts from the denoised-gridded image to generate the denoised, artifact-free gridded image, and wherein the PROPELLER reconstruction algorithm includes adjoint non-uniform fast Fourier transform blocks configured to grid the plurality of denoised blades into a Cartesian grid. In certain embodiments, the deep learning-based multi-level denoising network includes a phase filter disposed between the deep learning-based denoising model and the adjoint non-uniform fast Fourier transform blocks that is configured to perform parameter-less phase correction to correct for different phase in each of the blades. In certain embodiments, the deep learning-based artifact removing model has fewer parameters than the deep learning-based denoising model.

[0052] In certain embodiments, the processing component is configured to backpropagate a combination of both blade

level loss and grid level loss to train the deep learning-based multi-level denoising network from end to end. In certain embodiments, the combination of both the blade level loss and the grid level loss are backpropagated to train both the deep learning-based denoising model and the deep learning-based artifact removing model.

[0053] In certain embodiments, the processing component is configured to utilize the PROPELLER reconstruction algorithm to generate a noisy gridded image (e.g., via inverse two-dimensional (2D) fast Fourier transformation (FFT)) from the plurality of blades that have not been denoised. In certain embodiments, the processing component is configured to input both the denoised-gridded image and the noisy gridded image into the deep learning-based artifact removing model, wherein the deep learning-based artifact removing model utilizes both the denoised-gridded image and the noisy gridded image to generate the denoised, artifact-free gridded image. In certain embodiments, the processing component is configured to apply super-resolution to the denoised, artifact-free gridded image to generate a higher resolution denoised, artifact-free gridded image that is further denoised. In certain embodiments, the denoised, artifact-free gridded image may be subject to other post-processing. In certain embodiments, the processing component is configured to write the denoised, artifact-free gridded image to a DICOM file (e.g., for viewing or storage).

[0054] An additional interface circuit **172** may be provided for exchanging image data, configuration parameters, and so forth with external system components such as remote access and storage devices **108**. Finally, the system control and analysis circuit **168** may be communicatively coupled to various peripheral devices for facilitating operator interface and for producing hard copies of the reconstructed images. In the illustrated embodiment, these peripherals include a printer **174**, a monitor **176**, and user interface **178** including devices such as a keyboard, a mouse, a touchscreen (e.g., integrated with the monitor **176**), and so forth.

[0055] FIG. 2 is a schematic diagram illustrating a deep learning-based multi-level denoising network **180** configured to improve image quality of PROPELLER imaging. The deep learning-based multi-level denoising network **180** includes a deep learning-based denoising model **182** and a deep learning-based artifact removing model **184**. The deep learning-based multi-level denoising network **180** also includes a phase filter **186** and adjoint non-uniform fast Fourier transform (NUFFT) layers or blocks **188** disposed between the deep learning-based denoising model **182** and the deep learning-based artifact removing model **184**. The phase filter **186** is disposed between the deep learning-based denoising model **182** and the adjoint NUFFT layers or blocks **188**. Both the deep learning-based denoising model **182** and the deep learning-based artifact removing model **184** are trainable layers. Both the phase filter **186** and the adjoint NUFFT layers **188** are not trainable layers. The adjoint NUFFT layers **188** have no parameters.

[0056] As depicted, the deep learning-based denoising model **182** is a residual dense network model having 5 blocks. In certain embodiments, both the type of learning model and/or the number of blocks/layer/parameters may vary. The deep learning-based artifact removing model **184** is a residual dense network model having 2 blocks. As depicted, the deep learning-based artifact removing model **184** is light or lightweight deep learning-based model. The

deep learning-based artifact removing model **184** has fewer parameters than the deep learning-based denoising model **182** as it does not take as many parameters to remove artifacts. In certain embodiments, the deep learning-based artifact removing model **184** may be of the same size or larger than the deep learning-based denoising model **182**.

[0057] A plurality of blades **190** (e.g., noisy blades) in an image domain (e.g., derived from blades of k-space data via inverse 2D FFT) of a region of interest are input into the deep learning-based denoising model **182**. The plurality of blades of k-space data are acquired (e.g., from a single channel or multiple channels) in a rotational manner (e.g., rotated approximately 10 to 20 degrees between blade acquisitions) around a center of k-space via a magnetic resonance imaging (MRI) scanner (e.g., magnetic resonance imaging scanner **102** in FIG. 1) from a coil (e.g., RF receiving coil such as a body coil or surface coil) during a PROPELLER sequence. Each blade of the plurality of blades of k-space data includes a plurality of parallel phase encoding lines sampled in a phase encoding order (e.g., having a rectilinear shape) using fast spin echo or gradient echo methods (i.e., each blade **190** is filled by an echo train of a respective MR pulse sequence). The deep learning-based denoising model **182** is configured to denoise each blade **190** (e.g., individually) of the plurality of blades **190** in the image domain to generate a plurality of denoised blades **192** (e.g., noise free blades). The deep learning-based denoising model **182** is pre-trained or primed to predict noise present in a Cartesian acquisition image (e.g., MR image). The deep learning-based denoising model **182** is built to be size invariant. The training of the deep learning-based denoising model **182** is as described below in FIG. 6.

[0058] The denoised blades **192** are then inputted into the phase filter **186**. The phase filter **186** is configured to perform parameter-less phase correction to correct for different phase in each of the blades **192**. Performing phase correction for each denoised blade **192** assures its point of rotation is exactly at the center of k-space. After phase correction, the phase-corrected, denoised blades **192** are inputted into the adjoint NUFFT layers **188**. The adjoint NUFFT layers **188** are configured to grid the plurality of denoised blades **192** into a Cartesian grid to generate a denoised (noise free)-gridded image **194** (e.g., a first complex image). The adjoint NUFFT layers **188** are configured to perform corrections for bulk in-plane rotation and in-plane translation of the object and to perform correlation-weighting to minimize the data from blades containing motion or displacement errors.

[0059] In addition, the noisy blades **190** are inputted into the phase filter **186** as indicated by arrow **193**. The phase filter **186** is configured to perform parameter-less phase correction to correct for different phase in each of the blades **190**. After phase correction, the phase-corrected, noisy blades **190** are inputted into the adjoint NUFFT layers **188** as indicated by arrow **195**. The adjoint NUFFT layers **188** are configured to grid the plurality of noisy blades **190** into a Cartesian grid to generate a noisy gridded image **196** (e.g., a second complex image).

[0060] Both the denoised (noise free)-gridded image **194** and the noisy gridded image **196** are inputted into the deep learning-based artifact removing model **184**. The deep learning-based artifact removing model **184** is configured to remove individual-based denoising-induced artifacts from the denoised-gridded image **194** to generate a denoised,

artifact-free gridded image **198** utilizing both the denoised (noise free)-gridded image **194** and the noisy gridded image **196**.

[0061] Both blade level loss **200** (loss obtained after individual denoising of blades) and grid level loss **202** (loss obtained after gridding of blades) are combined with different respective weights given to the blade level loss **200** and the grid level loss **202**. The combined blade level loss **200** and the grid level loss **202** are backpropagated to train the deep learning-based multi-level denoising network **180** from end to end. In particular, the combination of both the blade level loss **200** and the grid level loss **202** are backpropagated to train both the deep learning-based denoising model **182** and the deep learning-based artifact removing model **184**.

[0062] FIG. 3 is a flow chart of a method **204** for improving image quality of PROPELLER imaging. One or more steps of the method **204** may be performed by processing circuitry of the magnetic resonance imaging system **100** in FIG. 1 or a remote computing system. For example, the processing circuitry may part of the scanner control circuitry **104** and/or system control circuitry **106** of the magnetic resonance imaging system **100**. One or more steps of the method **204** may be performed simultaneously and/or in a different order shown in FIG. 3. The method **204** is performed on data acquired from a single channel of a coil (e.g., RF receiving coil such as a body coil or surface coil).

[0063] The method **204** includes acquiring a plurality of blades of k-space data of a region of interest in a rotational manner (e.g., rotated approximately 10 to 20 degrees between blade acquisitions) around a center of k-space via a magnetic resonance imaging scanner (e.g., magnetic resonance imaging scanner **102** in FIG. 1) from a coil (e.g., RF receiving coil such as a body coil or surface coil) during a PROPELLER sequence (block **206**). Each blade of the plurality of blades of k-space data includes a plurality of parallel phase encoding lines sampled in a phase encoding order (e.g., having a rectilinear shape) using fast spin echo or gradient echo methods (i.e., each blade is filled by an echo train of a respective MR pulse sequence). The plurality of blades are converted to the image domain via inverse 2D FFT.

[0064] The method **204** also includes utilizing a deep learning-based multi-level denoising network to denoise each blade (e.g., individually) of the plurality of blades in an image domain to generate a plurality of denoised blades, to utilize a PROPELLER reconstruction algorithm to generate a denoised-gridded image from the plurality of denoised blades, and to remove individual-based denoising-induced artifacts from the denoised-gridded image to generate a denoised, artifact-free gridded image (block **208**). In particular, a deep learning-based denoising model to denoise each blade of the plurality of blades in the image domain to generate the plurality of denoised blades and a deep learning-based artifact removing model to remove the individual-based denoising-induced artifacts from the denoised-gridded image to generate the denoised, artifact-free gridded image. The deep learning-based denoising model is trained to predict noise present in a Cartesian acquisition image (e.g., MR image). In addition, the PROPELLER reconstruction algorithm includes adjoint non-uniform fast Fourier transform blocks configured to grid the plurality of denoised blades into a Cartesian grid. The PROPELLER reconstruction algorithm includes parameter-less phase correction for each blade to assure its point of rotation is exactly at the

center of k-space, corrections for bulk in-plane rotation and in-plane translation of the object, and correlation-weighting to minimize the data from blades containing motion or displacement errors. In certain embodiments, the denoised, artifact-free gridded image may be further processed (as described below in FIG. 5) prior to writing the image into a Digital Imaging and Communications in Medicine (DICOM) file.

[0065] FIG. 4 is a flow chart of a method **210** for improving image quality of PROPELLER imaging (e.g., utilizing data from multiple channels). One or more steps of the method **210** may be performed by processing circuitry of the magnetic resonance imaging system **100** in FIG. 1 or a remote computing system. For example, the processing circuitry may part of the scanner control circuitry **104** and/or system control circuitry **106** of the magnetic resonance imaging system **100**. One or more steps of the method **210** may be performed simultaneously and/or in a different order shown in FIG. 4. The method **210** is performed on data acquired from a plurality of channels of a coil (e.g., RF receiving coil such as a body coil).

[0066] The method **210** includes acquiring a plurality of blades of k-space data (e.g., from multiple channels) of a region of interest in a rotational manner (e.g., rotated approximately 10 to 20 degrees between blade acquisitions) around a center of k-space via a magnetic resonance imaging scanner (e.g., magnetic resonance imaging scanner **102** in FIG. 1) from a coil (e.g., RF receiving coil such as a body coil or surface coil) during a PROPELLER sequence (block **212**). Each blade of the plurality of blades of k-space data includes a plurality of parallel phase encoding lines sampled in a phase encoding order (e.g., having a rectilinear shape) using fast spin echo or gradient echo methods i.e., each blade is filled by an echo train of a respective MR pulse sequence). The method **210** also includes combining the plurality of blades of k-space data acquired from the plurality of channels prior to utilizing a deep learning-based multi-level denoising network to denoise each blade of the plurality of blades of k-space data (i.e., corresponding blades of k-space data from the channels are combined) (block **214**). The plurality of blades are converted to the image domain via inverse 2D FFT.

[0067] The method **210** also includes utilizing a deep learning-based multi-level denoising network to denoise (e.g., individually) each blade (e.g., combined blade) of the plurality of blades in the image domain to generate a plurality of denoised blades, to utilize a PROPELLER reconstruction algorithm to generate a denoised-gridded image from the plurality of denoised blades, and to remove individual-based denoising-induced artifacts from the denoised-gridded image to generate a denoised, artifact-free gridded image (block **216**). In particular, a deep learning-based denoising model to denoise each blade of the plurality of blades in the image domain to generate the plurality of denoised blades and a deep learning-based artifact removing model to remove the individual-based denoising-induced artifacts from the denoised-gridded image to generate the denoised, artifact-free gridded image. The deep learning-based denoising model is trained to predict noise present in a Cartesian acquisition image (e.g., MR image). In addition, the PROPELLER reconstruction algorithm includes adjoint non-uniform fast Fourier transform blocks configured to grid the plurality of denoised blades into a Cartesian grid. The PROPELLER reconstruction algorithm includes parameter-

less phase correction for each blade to assure its point of rotation is exactly at the center of k-space, corrections for bulk in-plane rotation and in-plane translation of the object, and correlation-weighting to minimize the data from blades containing motion or displacement errors. In certain embodiments, the denoised, artifact-free gridded image may be further processed (as described below in FIG. 5) prior to writing the image into a Digital Imaging and Communications in Medicine (DICOM) file.

[0068] FIG. 5 is a flow chart of a method **218** for improving image quality of PROPELLER imaging (e.g., with details of PROPELLER reconstruction). One or more steps of the method **218** may be performed by processing circuitry of the magnetic resonance imaging system **100** in FIG. 1 or a remote computing system. For example, the processing circuitry may part of the scanner control circuitry **104** and/or system control circuitry **106** of the magnetic resonance imaging system **100**. One or more steps of the method **218** may be performed simultaneously and/or in a different order shown in FIG. 5. The method **218** is performed on data acquired from either a single channel (e.g., as described in the method **204** in FIG. 3) or a plurality of channels (e.g., as described in the method **210** in FIG. 4) of a coil (e.g., RF receiving coil such as a body coil or surface coil).

[0069] The method **218** includes acquiring a plurality of blades (e.g., noisy blades) of k-space data (e.g., from a single channel or multiple channels) of a region of interest in a rotational manner (e.g., rotated approximately **10** to **20** degrees between blade acquisitions) around a center of k-space via a magnetic resonance imaging scanner (e.g., magnetic resonance imaging scanner **102** in FIG. 1) from a coil (e.g., RF receiving coil such as a body coil or surface coil) during a PROPELLER sequence (block **220**). Each blade of the plurality of blades of k-space data includes a plurality of parallel phase encoding lines sampled in a phase encoding order (e.g., having a rectilinear shape) using fast spin echo or gradient echo methods (i.e., each blade is filled by an echo train of a respective MR pulse sequence). The plurality of blades are converted to the image domain via inverse 2D FFT

[0070] The method **218** also includes utilizing a deep learning-based denoising model of a deep learning-based multi-level denoising network to denoise each blade (or each combined blade) of the plurality of blades in the image domain to generate a plurality of denoised blades (block **222**). The deep learning-based denoising network is trained to predict noise present in a Cartesian acquisition image (e.g., MR image). The deep learning-based denoising network is built to be size invariant. The training of the deep learning-based denoising network is described below in FIG. 6.

[0071] As noted above, the PROPELLER reconstruction algorithm includes parameter-less phase correction for each blade to assure its point of rotation is exactly at the center of k-space, corrections for bulk in-plane rotation and in-plane translation of the object, and correlation-weighting to minimize the data from blades containing motion or displacement errors. The method **218** further includes performing parameter-less phase correction, via a phase filter, on the plurality of denoised blades (block **224**). Performing phase correction for each blade assures its point of rotation is exactly at the center of k-space. The method **218** still further includes determining the weightage for each blade and estimating sampling density correction for each blade (block

226). Based on the weights and the estimated sampling density corrections to be applied to the denoised blades, the method **218** includes gridding, via an adjoint NUFFT, all of the denoised blades of k-space into a Cartesian matrix (e.g., uniform k-space co-ordinates) to generate a denoised, gridded image with artifacts (block **228**). The blocks **224** and **226** of the method **218** ensure corrections for bulk in-plane rotation and in-plane translation of the object, while the correlation-weighting minimizes the data from the blades containing motion or displacement errors.

[0072] The method **218** further includes performing parameter-less phase correction, via a phase filter, on plurality of blades (noisy blades) (block **230**). The method **218** still further includes determining the weightage for each noisy blade and estimating sampling density correction for each noisy blade (block **232**). Based on the weights and the estimated sampling density corrections to be applied to the noisy blades, the method **218** includes gridding, via an adjoint NUFFT, all of the noisy blades of k-space into a Cartesian matrix (e.g., uniform k-space co-ordinates) to generate a noisy gridded image (block **234**).

[0073] The method **218** includes utilizing a deep learning-based artifact removing model to remove the individual-based denoising-induced artifacts from the denoised-gridded image to generate a denoised, artifact-free gridded image (block **236**). Both the denoised (noise free)-gridded image and the noisy gridded image are inputted into the deep learning-based artifact removing model. The deep learning-based artifact removing model is configured to remove individual-based denoising-induced artifacts from the denoised-gridded image to generate a denoised, artifact-free gridded image utilizing both the denoised (noise free)-gridded image and the noisy gridded image.

[0074] In certain embodiments, the method **218** also includes applying super-resolution to the denoised, artifact-free gridded image to generate a higher resolution denoised, artifact-free gridded image that is further denoised (e.g., free of blur) (block **238**). In certain embodiments, the denoised, artifact-free gridded image may be subject to other post-processing. The method **218** further includes writing the image (e.g., denoised, artifact-free gridded image or higher resolution denoised, artifact-free gridded image) to a DICOM file (e.g., for viewing or storage) (block **240**).

[0075] The method **218** even further includes backpropagating a combination of both blade level loss and grid level loss to train the deep learning-based multi-level denoising network from end to end (block **242**). The combination of both the blade level loss and the grid level loss are back-propagated to train both the deep learning-based denoising model and the deep learning-based artifact removing model.

[0076] FIG. 6 illustrates a flow chart of a method **244** for training (e.g., initial training) a deep learning-based denoising model **182**. As noted above, the deep learning-based denoising model is primed or trained prior to be utilized with the deep learning-based multi-level denoising network. One or more steps of the method **244** may be performed by processing circuitry of the magnetic resonance imaging system **100** in FIG. 1 or a remote computing system. For example, the processing circuitry may part of the scanner control circuitry **104** and/or system control circuitry **106** of the magnetic resonance imaging system **100**.

[0077] The method **244** includes inputting input-output (e.g., labeled) data pairs **246** into a neural network **248** (block **249**). In certain embodiments, the neural network **248**

is a residual dense network. In certain embodiments, input-output data pairs **246** include near perfect (i.e., sharp, lacking artifacts, and high signal-to-noise ratio) and conventional MRI like images **250** simulated from natural images **252**. The method **244** also includes training the neural network **248** on the input-output data pairs **246** utilizing supervised learning to generate the deep learning-based denoising model **182** (block **253**). In certain embodiments, at least some pairs of the input-output data pairs **246** include skewed aspect ratios to augment training to improve the performance of the deep learning-based denoising model **182**. The deep learning-based denoising model **182** is trained to predict noise in Cartesian acquired images (e.g., MR images). Due to the training on natural images with sufficient variability and the residue to be predicted remains in the same Gaussian distribution, the deep learning-based can be utilized on blade level images. In certain embodiments, the images of the input-output data pairs **246** may be 256×256 . However, the deep learning-based denoising model **182** is built to be size invariant.

[0078] FIGS. 7-11 illustrate the effectiveness of performing multi-level denoising. FIG. 7 depicts PROPELLER images of a spine comparing multi-level denoising and grid level denoising. Images **254**, **256**, and **258** were acquired of a spine (with a sagittal T2 scan) utilizing PROPELLER with a 1.5 Tesla (T) magnetic resonance imaging scanner using a body coil. Image **254** is a PROPELLER image without denoising. Image **256** is the corresponding PROPELLER image with multi-level denoising performed utilizing the deep learning-based multi-level denoising network described above. Image **258** is the corresponding PROPELLER image with grid level denoising performed (i.e., denoising performed after the gridding of the blades). Both images **256** and **258** are 100 percent denoised. The quality (e.g., signal-to-noise ratio and sharpness) is improved in images **256** and **258** compared to the image **254**. However, image **258** has patchy denoising in some regions, for example, when looking at respective zoomed regions **260**, **262**, and **264** of the images **254**, **256**, and **258** as seen in the zoomed region **264**.

[0079] FIG. 8 depicts PROPELLER images of a knee comparing multi-level denoising and grid level denoising. Images **266**, **268**, and **270** were acquired of a knee (with a sagittal proton density-weighted fat-suppressed scan) utilizing PROPELLER with a 1.5 Tesla (T) magnetic resonance imaging scanner using a body coil. Image **266** is a PROPELLER image without denoising. Image **268** is the corresponding PROPELLER image with multi-level denoising performed utilizing the deep learning-based multi-level denoising network described above. Image **270** is the corresponding PROPELLER image with grid level denoising performed (i.e., denoising performed after the gridding of the blades). Both images **268** and **270** are 100 percent denoised. The quality (e.g., signal-to-noise ratio and sharpness) is improved in images **268** and **270** compared to the image **266**. However, the model utilized for grid level denoising has exaggerated certain structures as seen in the image **270**.

[0080] FIG. 9 depicts additional PROPELLER images of a knee comparing multi-level denoising and grid level denoising. Images **272**, **274**, and **276** were acquired of a knee (with a sagittal T2 scan) utilizing PROPELLER with a 1.5 Tesla (T) magnetic resonance imaging scanner using a body coil. Image **272** is a PROPELLER image without

denoising. Image **274** is the corresponding PROPELLER image with multi-level denoising performed utilizing the deep learning-based multi-level denoising network described above. Image **276** is the corresponding PROPELLER image with grid level denoising performed (i.e., denoising performed after the gridding of the blades). Both images **274** and **276** are 100 percent denoised. The quality (e.g., signal-to-noise ratio and sharpness) is improved in images **274** and **276** compared to the image **272**. However, the model utilized for grid level denoising has wiped/blurred out some structures when looking at respective zoomed regions **278**, **280**, and **282** of the images **272**, **274**, and **276** as seen in the zoomed region **282**.

[0081] FIG. 10 depicts PROPELLER images of a spine comparing multi-level denoising and grid level denoising with super-resolution applied. Images **284**, **286**, and **288** were acquired of a spine (with a T2 scan) utilizing PROPELLER with a 0.5 Tesla (T) magnetic resonance imaging scanner (which generates low signal-to-noise ratio data) using a neck array coil. Image **284** is a PROPELLER image without denoising. Super-resolution was applied to generate the images **286** and **288**. Image **286** is the corresponding PROPELLER image with multi-level denoising performed utilizing the deep learning-based multi-level denoising network described above. Image **288** is the corresponding PROPELLER image with grid level denoising performed (i.e., denoising performed after the gridding of the blades). Both images **286** and **288** are 100 percent denoised. The quality (e.g., signal-to-noise ratio and sharpness) is improved in images **286** and **288** compared to the image **284**. In addition, the signal-to-noise ratio is decent in both images **286** and **288**.

[0082] FIG. 11 depicts PROPELLER images of a pelvis comparing blade level denoising and multi-level denoising. Images **290**, **292**, and **294** were acquired of a pelvis (with a sagittal T2 scan) utilizing PROPELLER with a 1.5 Tesla (T) magnetic resonance imaging scanner using a body coil. Image **290** is a PROPELLER image without denoising. Image **292** is the corresponding PROPELLER image with blade level denoising performed utilizing the deep learning-based blade level denoising model (i.e., but not utilizing a deep learning-based artifact removing model to remove the individual-based denoising-induced artifacts). Image **294** is the corresponding PROPELLER image with multi-level denoising performed utilizing the deep learning-based multi-level denoising network described above. Both images **292** and **294** are 100 percent denoised. The quality (e.g., signal-to-noise ratio and sharpness) is improved in images **292** and **294** compared to the image **290**. Comparing image **294** to image **292**, it can be observed that the multi-level denoising network resolves structures much sharper than the blade level denoising model.

[0083] To evaluate the performance of multi-level denoising relative to grid level denoising single channel PROPELLER images were simulated from natural images. FIG. 12 depicts an example of a PROPELLER ground truth image **296** of a natural image. From the natural image, 19 blades were simulated of dimension 322×32 and gridded together to form the PROPELLER ground truth image **296**. Different levels of random Gaussian were added to the blades and gridded to obtain a noisy PROPELLER image. Denoising was performed using grid level denoising and multi-level denoising techniques.

[0084] The results are depicted in FIG. 13. FIG. 13 depicts a graph 298 analyzing performance of grid level denoising relative to multi-level denoising on a natural image. The graph 298 includes an x-axis 300 representing noise level and a y-axis 302 representing structural similarity index (SSIM). Plot 304 represents the SSIM between a ground truth PROPELLER image and PROPELLER images derived utilizing the grid level denoising technique on noisy PROPELLER images. Plot 306 represents the SSIM between a ground truth PROPELLER image and PROPELLER images derived utilizing the multi-level denoising technique on noisy PROPELLER images. It is observed, as depicted in the graph 298, that as the noise added increases, the ability of the noise to completely morph or bury the structures after gridding increases and hence, it becomes impossible for grid level denoising to recover those structures. On the other hand, as the multi-level denoising model attempts to remove noise at blade level, where noise remains pixelwise statistically independent Gaussian noise, the possibility of recovering those structures is high.

[0085] FIG. 14 depicts examples of images from the analysis of simulated PROPELLER images of a natural image in FIG. 13. Images 308 and 310 are the same ground truth PROPELLER image. Images 312 and 314 are noisy PROPELLER images simulated with different noise levels, respectively. Images 316 and 318 are the corresponding PROPELLER images to the images 312 and 314, respectively, obtained by multi-level denoising. Images 320 and 322 are the corresponding PROPELLER images to the images 312 and 314, respectively, obtained by grid level denoising. As noted above, the multi-level denoising is better at recovering the structures.

[0086] Technical effects of the disclosed subject matter include providing system and methods for improving image quality of periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) imaging. In particular, the present disclosure provides for denoising individual blades in image domain prior to phase correcting and gridding the blades (where the noise is still Gaussian). Technical effects of the disclosed subject matter include providing a deep learning-based multi-level denoising network that provides an end to end trainable deep learning framework completely immersive into PROPELLER reconstruction that is configured to achieve denoising with high fidelity and minimal artifacts. Technical effects of the disclosed subject matter include improve image quality in with regard to both signal-to-noise ratio and sharpness.

[0087] The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

[0088] This written description uses examples to disclose the present subject matter, including the best mode, and also to enable any person skilled in the art to practice the subject matter, including making and using any devices or systems and performing any incorporated methods. The patentable

scope of the subject matter is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

1. A computer-implemented method for improving image quality of periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) imaging, comprising:

acquiring, via a processor, a plurality of blades of k-space data of a region of interest in a rotational manner around a center of k-space via a magnetic resonance imaging (MRI) scanner from a coil during a PROPELLER sequence, wherein each blade of the plurality of blades of k-space data comprises a plurality of parallel phase encoding lines sampled in a phase encoding order; and

utilizing, via the processor, a deep learning-based multi-level denoising network to denoise each blade of the plurality of blades in an image domain to generate a plurality of denoised blades, to utilize a PROPELLER reconstruction algorithm to generate a denoised-gridded image from the plurality of denoised blades, and to remove individual-based denoising-induced artifacts from the denoised-gridded image to generate a denoised, artifact-free gridded image.

2. The computer-implemented method of claim 1, wherein the deep learning-based multi-level denoising network comprises a deep learning-based denoising model to denoise each blade of the plurality of blades in the image domain to generate the plurality of denoised blades and a deep learning-based artifact removing model to remove the individual-based denoising-induced artifacts from the denoised-gridded image to generate the denoised, artifact-free gridded image, and wherein the PROPELLER reconstruction algorithm comprises adjoint non-uniform fast Fourier transform blocks configured to grid the plurality of denoised blades into a Cartesian grid.

3. The computer-implemented method of claim 2, wherein the deep learning-based artifact removing model has fewer parameters than the deep learning-based denoising model.

4. The computer-implemented method of claim 2, further comprising backpropagating, via the processor, a combination of both blade level loss and grid level loss to train the deep learning-based multi-level denoising network from end to end.

5. The computer-implemented method of claim 4, wherein the combination of both the blade level loss and the grid level loss are backpropagated to train both the deep learning-based denoising model and the deep learning-based artifact removing model.

6. The computer-implemented method of claim 2, further comprising utilizing, via the processor, the PROPELLER reconstruction algorithm to generate a noisy gridded image from the plurality of blades that have not been denoised.

7. The computer-implemented method of claim 6, further comprising inputting, via the processor, both the denoised-gridded image and the noisy gridded image into the deep learning-based artifact removing model, wherein the deep learning-based artifact removing model utilizes both the

denoised-gridded image and the noisy gridded image to generate the denoised, artifact-free gridded image.

8. The computer-implemented method of claim 1, further comprising applying, via the processor, super-resolution to the denoised, artifact-free gridded image to generate a higher resolution denoised, artifact-free gridded image that is further denoised.

9. A system for improving image quality of periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) imaging, comprising:

a memory encoding processor-executable routines; and
a processor configured to access the memory and to execute the processor-executable routines, wherein the processor-executable routines, when executed by the processor, cause the processor to:

acquire a plurality of blades of k-space data of a region of interest in a rotational manner around a center of k-space via a magnetic resonance imaging (MRI) scanner from a coil during a PROPELLER sequence, wherein each blade of the plurality of blades of k-space data comprises a plurality of parallel phase encoding lines sampled in a phase encoding order; and

utilize a deep learning-based multi-level denoising network to denoise each blade of the plurality of blades in an image domain to generate a plurality of denoised blades, to utilize a PROPELLER reconstruction algorithm to generate a denoised-gridded image from the plurality of denoised blades, and to remove individual-based denoising-induced artifacts from the denoised-gridded image to generate a denoised, artifact-free gridded image.

10. The system of claim 9, wherein the deep learning-based multi-level denoising network comprises a deep learning-based denoising model to denoise each blade of the plurality of blades in the image domain to generate the plurality of denoised blades and a deep learning-based artifact removing model to remove the individual-based denoising-induced artifacts from the denoised-gridded image to generate the denoised, artifact-free gridded image, and wherein the PROPELLER reconstruction algorithm comprises an adjoint non-uniform fast Fourier transform blocks configured to grid the plurality of denoised blades into a Cartesian grid.

11. The system of claim 10, wherein the deep learning-based artifact removing model has fewer parameters than the deep learning-based denoising model.

12. The system of claim 10, wherein the processor-executable routines, when executed by the processor, further cause the processor to backpropagate a combination of both blade level loss and grid level loss to train the deep learning-based multi-level denoising network from end to end.

13. The system of claim 12, wherein the combination of both the blade level loss and the grid level loss are backpropagated to train both the deep learning-based denoising model and the deep learning-based artifact removing model.

14. The system of claim 10, wherein the processor-executable routines, when executed by the processor, further cause the processor to utilize the PROPELLER reconstruction algorithm to generate a noisy gridded image from the plurality of blades that have not been denoised.

15. The system of claim 14, wherein the processor-executable routines, when executed by the processor, further cause the processor to input both the denoised-gridded image and the noisy gridded image into the deep learning-based artifact removing model, wherein the deep learning-based artifact removing model utilizes both the denoised-gridded image and the noisy gridded image to generate the denoised, artifact-free gridded image.

16. The system of claim 9, wherein the processor-executable routines, when executed by the processor, further cause the processor to apply super-resolution to the denoised, artifact-free gridded image to generate a higher resolution denoised, artifact-free gridded image that is further denoised.

17. A non-transitory computer-readable medium, the computer-readable medium comprising processor-executable code that when executed by a processor, causes the processor to:

acquire a plurality of blades of k-space data of a region of interest in a rotational manner around a center of k-space via a magnetic resonance imaging (MRI) scanner from a coil during a periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) sequence, wherein each blade of the plurality of blades of k-space data comprises a plurality of parallel phase encoding lines sampled in a phase encoding order; and

utilize a deep learning-based multi-level denoising network to denoise each blade of the plurality of blades in an image domain to generate a plurality of denoised blades, to utilize a PROPELLER reconstruction algorithm to generate a denoised-gridded image from the plurality of denoised blades, and to remove individual-based denoising-induced artifacts from the denoised-gridded image to generate a denoised, artifact-free gridded image.

18. The computer-readable medium of claim 17, wherein the deep learning-based multi-level denoising network comprises a deep learning-based denoising model to denoise each blade of the plurality of blades in the image domain to generate the plurality of denoised blades and a deep learning-based artifact removing model to remove the individual-based denoising-induced artifacts from the denoised-gridded image to generate the denoised, artifact-free gridded image, and wherein the PROPELLER reconstruction algorithm comprises an adjoint non-uniform fast Fourier transform blocks configured to grid the plurality of denoised blades into a Cartesian grid.

19. The computer-readable medium of claim 18, wherein the processor-executable code, when executed by the processor, further causes the processor to further backpropagate a combination of both blade level loss and grid level loss to train the deep learning-based multi-level denoising network from end to end.

20. The computer-readable medium of claim 19, wherein the combination of both the blade level loss and the grid level loss are backpropagated to train both the deep learning-based denoising model and the deep learning-based artifact removing model.

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